

LIAM NANCE

Aerospace Engineer | U.S Citizen

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EDUCATION

Arizona State University

MS - Aerospace Engineering: GPA 4.00/4.00, *With Distinction*

May 2026

BSE - Aerospace Engineering: GPA: 3.91/4.00, *Summa Cum Laude*

May 2025

SKILLS

Engineering Software: MATLAB, Python, Siemens NX, SolidWorks, Fusion 360, Ansys Fluent (CFD), XFOIL, Arduino, AutoCAD

Technical Domains: GD&T (ASME Y14.5-2009), CAD, CFD, Aerodynamics, Propulsion, Compressible Flow, Thermodynamics

Manufacturing: CNC Router, CNC Mill, Laser Cutting, Toolpathing, Vcarve, Solumina/MES

PROGRAMS AND PROJECTS

MATLAB Proportional Navigation Algorithm

May, 2026

- Built to learn guidance algorithm fundamentals. Employed TPN and PPN schemes for an intercept missile tracking a hypersonic target.
- Integrated a Kalman filter to smooth and predict noisy data—tuned the controller and filter to ensure a successful intercept.

Serial Robotic Arm Control Scheme

Apr, 2026

- Worked in a team to develop a program to control serial robotic arms of any number of joints and links.
- Integrated PD compliance control, impedance control, forward and inverse kinematics, and dynamics.
- Created custom GUI for analysis of motion, forces, and controller tuning.

Aphrodite 1 - Theoretical Rotating Space Station

Aug, 2024 - May, 2025

- Thermal Analysis lead. Designed a collapsible radiator system integrated with internal fluid loops and heat exchangers.
- Designed an 8-DOF robotic arm for in-orbit assembly and maintenance using CAD and 3D printing.
- Performed materials and coatings trade studies based on thermal emissivity, longevity, and MMOD resistance.

Thorough Analysis of Ramjet and Scramjet Flow - MATLAB

Dec, 2025

- Developed a MATLAB-based performance tool for non-isentropic ram-compression Brayton Cycle engines.
- Enabled parametric studies on Mach number, flight altitude, internal flow speeds, maximum material temperatures, isentropic efficiencies, pressures, ISP, and TSFC for engine optimization.

Spaceworks at ASU - NASA JPL Partner Program

Aug, 2024 - Aug, 2025

- Applied the NASA standard procedure and documents in a professional team setting. Visited the JPL Campus.
- Lead engineer and designer for the construction of an "Earth Lander" which dropped from 100+ feet, and successfully delivered a payload to the ground. Used CAD and CFD for systems design, in addition to physical modeling.

Rigorous Analysis of Liquid Rocket Engine (LRE)

Dec, 2024

- Calculated flow properties of LOX/LH2 at 15+engine locations, including preburners, turbines, injectors, and nozzle.
- Identified key engine parameters such as thrust, temperature limit, maximum flow velocity, and engine efficiency.

Boy Scouts of America - Eagle Scout

WORK EXPERIENCE

Northrop Grumman Corporation

Chandler, AZ

Manufacturing Engineer 2, Thermal Protection Systems

Aug, 2026 - Present

- Create designs, toolpathing, and processes for installing and improving the thermal protection on space vehicles.

Chas Roberts Plumbing and AC

Peoria, AZ

CAD Drafter

Aug, 2024 - Aug, 2026

- Design detailed AutoCAD HVAC drawings containing hundreds of components, adhering to engineering specifications.
- Review technical drawings and specification sheets to ensure proper system integration from design through implementation.

Greystar Real Estate Partners

Tempe, AZ

Leasing Assistant

Mar, 2023 - Jun, 2024

- Learned to communicate in a team environment and facilitated over \$500,000 in leases for the property.

Microsoft

Tempe, AZ

Support Technician - SMB

Feb, 2022 - Mar, 2023

- Managed high-stakes support cases—resolved complex software issues using root cause analysis